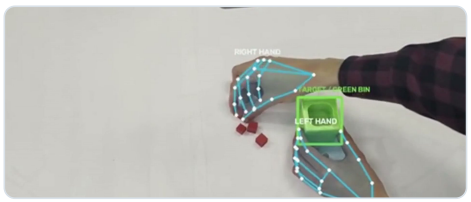
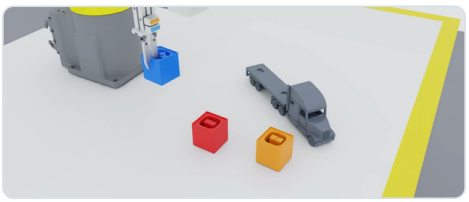
Adaptive Workcell Reconfiguration

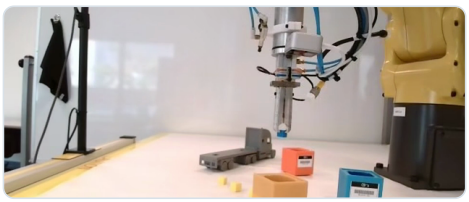
We built a physical-AI workflow for high-mix assembly: capture intent from a human demonstration or digital-twin move, ground it in measured geometry, validate it in simulation, and execute a supervised plan on a FANUC ER-4iA.



Human demonstration



Isaac Sim validation



CSI FANUC execution

MANUFACTURING PROBLEM AND GOAL

Modern manufacturing must respond to frequent shifts in supply, demand, product mix, and global supply-chain disruptions. Yet adapting a robot cell to a new SKU can still take weeks or months of CAD cleanup, tool selection, point teaching, collision checks, and first-piece debugging. We built a reusable workflow that captures task intent once, then turns it into geometry and motion engineers can inspect, revise, and reuse. It is designed to move production reconfiguration from months toward days.

HOW THE PROJECT USES PHYSICAL AI

We turn multi-view demonstrations and digital-twin moves into hand and part tracks, contact points, and robot steps. Measured geometry defines where the gripper can approach safely and how parts fit together. We test candidate motions in NVIDIA Isaac Sim/PhysX, then use calibrated table, side, and wrist RGB-D cameras with robot state data to guide supervised execution. Engineers can inspect and edit every approach, grasp, transfer, insert, verification, and recovery step.

HOW WE USED CSI EQUIPMENT

At UWM's Connected Systems Institute, we calibrated and operated the FANUC ER-4iA with table, side, and wrist RGB-D cameras. Tests progressed from cube stacking and screw sorting to cube/bin/truck loading, orbit capture, and assembly of two-gear, four-gear, and inline-four printed demonstrators. Real hardware exposed scale, table-depth, finger-clearance, calibration, and fine-motion errors that simulation could not. We used those findings to improve the CAD, calibration, and recovery logic.

LIMITS AND NEXT VALIDATION

Performance depends on calibration and camera depth quality. Assemblies requiring 0.5 mm precision need sharper depth sensing and tighter calibration. More physical trials across varied parts and layouts will strengthen and validate the system.

DATA COLLECTED AND GENERATED

Our project archive documents 10 approved robot access runs from Oct. 20-31. Seven complete three-camera datasets contain 23 recordings, 142,335 timestamped robot state and action records, 237.5 minutes of robot operation, and 69 table, side, and wrist videos (11.9 camera-hours). Each record links joint and gripper state, tool position, robot commands, task timing, and synchronized camera views. Three additional runs retain requests, plans, logs, or snapshots.

Generated: annotated demonstrations, calibration and depth products, parametric robot-ready CAD, randomized Isaac Sim scenes, benchmark logs, and reference meshes.

EVIDENCE THAT THE APPROACH WORKS

On the physical FANUC, the organizer-scored bin-lift test achieved a successful lift. A later supervised run completed four cube-to-bin transfers in a row. We also recorded loaded-bin transport and multi-part assembly, with multi-view video and robot telemetry capturing each result.

In Isaac Sim, the final controller completed all six randomized truck-loading trials, improving from three of six. Every successful run placed all 12 cubes into three bins and loaded all three bins onto the truck. These results show a clear path from demonstration to repeatable robot execution.

EXPLORE THE TOOLKIT
theateam.dev

10-CELL ANNUAL CAPACITY MODEL

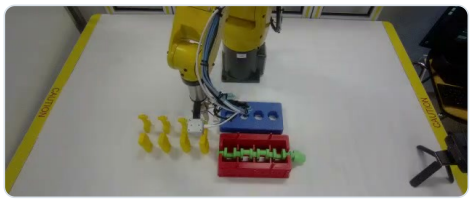
$490.5K
gross capacity / year



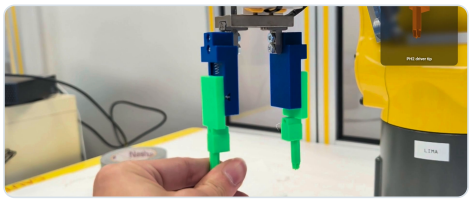
ENGINEERING 31%
$153K · 1,800 h

CELL CAPACITY 69%
$337.5K · 225 h

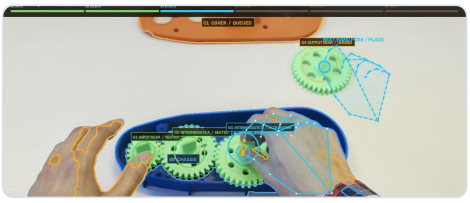
180 annual changeovers · gross value before implementation cost



Physical engine assembly



Interchangeable tool fingers



Video-derived assembly trace

The A-Team - Abdallah Benelmadjat, Pablo Camarillo, Benjamin Hess, Tar Meh, Nathan Baker
UWM CSI / MSOE / Rockwell Automation / NVIDIA